This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Review activities: MI, IS, WC, MU, AT, 82, NSA
User activities: SH, MC, GL
CODE C

DD FORM 1 SEP 67 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

P.A. ME
Other Cust AIR FORCE - 82

PROCUREMENT SPECIFICATION FF-S-107

SUPERSEDES: MS24649 AND AN 504

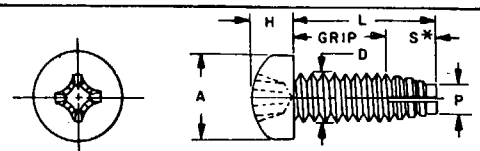
TITLE SCREW, TAPPING-THREAD CUTTING, TYPES D, F, G OR T, PAN HEAD, CROSS RECESSED, STEEL, CADMIUM PLATED

MILITARY STANDARD

MS24629

SHEET 1 OF 2

ENTIRE STANDARD REVISED



TYPE D

OPTIONAL THREAD TYPE



TYPE F TYPE G TYPE T

NOMINAL SIZE		2(.086)		4(.112)		6(.138)		8(.164)		10(.190)		1/4		10(.190)		1/4	
THREADS PER INCH		56		40		32		32		24		20		32		28	
MAJOR DIAMETER D	MAX	.0860		.1120		.1380		.1640		.1900		.2500		.1900		.2500	
	MIN	.0820		.1072		.1326		.1586		.1834		.2428		.1846		.2438	
HEAD DIAMETER A	MAX	.167		.219		.270		.322		.373		.492		.373		.492	
	MIN	.155		.205		.256		.306		.357		.473		.357		.473	
HEAD HEIGHT H	MAX	.062		.080		.097		.115		.133		.175		.133		.175	
	MIN	.053		.070		.087		.105		.122		.162		.122		.162	
POINT DIAMETER P	MAX	.067		.086		.106		.132		.147		.198		.158		.213	
	MIN	.061		.077		.095		.121		.133		.181		.147		.201	
LENGTH L	TOLERANCE	DASH NO.	GRIP MIN.	DASH NO.	GRIP MIN.	DASH NO.	GRIP MIN.	DASH NO.	GRIP MIN.	DASH NO.	GRIP MIN.	DASH NO.	GRIP MIN.	DASH NO.	GRIP MIN.	DASH NO.	GRIP MIN.
1/8	+0 -1/32	1	.031														
3/16		2	.076	9	.069	20	.047										
1/4		3	.138	10	.106	21	.109	33	.110								
5/16		4	.201	11	.169	22	.141	34	.141	44	.135			64	.135		
3/8		5	.263	12	.231	23	.203	35	.203	45	.156	56	.169	65	.156		
1/2		6	.388	13	.356	24	.328	36	.328	46	.281	57	.244	66	.281	73	.244
5/8		7	.513	14	.481	25	.453	37	.453	47	.406	58	.369	67	.406	74	.369
3/4				15	.606	26	.578	38	.578	48	.531	59	.494	68	.531	75	.494
7/8	+0 -3/64			16	.716	27	.688	39	.688	49	.641	60	.603	69	.641	76	.603
1				17	.841	28	.813	40	.813	50	.766	61	.728	70	.766	77	.728
1-1/4								41	1.063	51	1.016	62	.978	71	1.016	78	.978
1-1/2						30	1.313			52	1.266	63	1.228	72	1.266	79	1.228

MATERIAL: STEEL, CARBON, IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.
FINISH: CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3.
THREADS: THREAD TYPES TO BE IN ACCORDANCE WITH APPLICABLE REQUIREMENTS OF PROCUREMENT SPECIFICATION.
HEAT TREATMENT: CASE HARDENED OR HEAT TREATMENT TO MEET PERFORMANCE REQUIREMENTS OF PROCUREMENT SPECIFICATION.
RECESS: THE RECESS SHALL BE IN ACCORDANCE WITH MS9006.
HOLE SIZES: FOR APPROXIMATE HOLE SIZE SEE MS24634. FOR 1/4-28 SEE ASA B18.6.4.
DIMENSIONS: ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.
PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS NUMBER PLUS THE DASH NUMBER. EXAMPLE: MS24629-1.
*S (TAPER LENGTH) = 2-1/2 TO 3-1/2 TIMES THE PITCH DISTANCE FOR SCREWS HAVING A THREAD LENGTH OF 8 PITCHES OR LESS. TAPER LENGTH FOR LONGER LENGTHS WILL BE 3-1/2 TO 4-1/2 TIMES THE PITCH DISTANCE.
NOTES: (1) TYPE DESIGNATIONS ARE ESTABLISHED BY THE PROCUREMENT SPECIFICATION.
(2) FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
(3) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
(4) FOR AERONAUTICAL APPLICATIONS, THESE SCREWS ARE SUBJECT TO MS33749 LIMITATION AND INSTALLATION REQUIREMENTS.

FED. SUP CLASS 5305

APPROVED 9 OCTOBER 1959 REVISED (B) 25 FEBRUARY 1963 (C) 28 AUGUST 1967

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FED. SUP CLASS
5305

THE SCREWS COVERED BY THE FOLLOWING DASH NUMBERS LISTED IN MS24649 AND AN504 ARE CANCELLED AFTER 25 FEB 1963. USE THE DASH NUMBERS GIVEN IN SHEET I. THE CANCELLED SCREWS CANNOT ALWAYS REPLACE THE SUPERSEDING SCREWS AND SHOULD BE USED UNTIL EXISTING STOCKS ARE DEPLETED. USE ONLY THE SUPERSEDING SCREWS FOR DESIGN AND ENGINEERING. REPLACEMENT SHALL BE IN ACCORDANCE WITH THE TABLE BELOW.

CROSS REFERENCE			
CANCELLED			SUPERSEDED BY
MS 24649	AN 504		MS24629
DASH NO.	DASH NO.	DASH NO.	DASH NO.
1			2
2			3
3			4
10			9
11	4-4	4R4	10
12	4-5	4R5	11
13	4-6	4R6	12
14	4-8	4R8	13
15	4-10	4R10	14
16	4-12	4R12	15
21			20
22	6-4	6R4	21
23	6-5	6R5	22
24	6-6	6R6	23
25	6-8	6R8	24
26	6-10	6R10	25
27	6-12	6R12	26
28	6-14	6R14	27
29	6-16	6R16	28
32	8-4	8R4	33
33	8-5	8R5	34
34	8-6	8R6	35
35	8-8	8R8	36
36	8-10	8R10	37
37	8-12	8R12	38
38	8-14	8R14	39
39	8-16	8R16	40
	8-20	8R20	41
44	10-5	10R5	44
45	10-6	10R6	45
46	10-8	10R8	46
47	10-10	10R10	47
48	10-12	10R12	48
49	10-14	10R14	49
50	10-16	10R16	50
51	10-20	10R20	51
52	10-24	10R24	52
55			56
56	416-8	416R8	57
57	416-10	416R10	58
58	416-12	416R12	59
59	416-14	416R14	60
60	416-16	416R16	61
61	416-20	416R20	62
62	416-24	416R24	63

CROSS REFERENCE			
CANCELLED			SUPERSEDED BY
MS24649	AN 504		MS 24629
DASH NO.	DASH NO.	DASH NO.	DASH NO.
	1032-5	1032R5	64
	1032-6	1032R6	65
	1032-8	1032R8	66
	1032-10	1032R10	67
	1032-12	1032R12	68
	1032-14	1032R14	69
	1032-16	1032R16	70
	1032-20	1032R20	71
	1032-24	1032R24	72
	428-8	428R8	73
	428-10	428R10	74
	428-12	428R12	75
	428-14	428R14	76
	428-16	428R16	77
	428-20	428R20	78
	428-24	428R24	79

APPROVED 5 OCTOBER 1959 REVISED (B) 25 FEBRUARY 1963 (C) SEE SHEET I AND 2 FOR CHANGES

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PROCUREMENT SPECIFICATION FF–S–107	SUPERSEDES: MS 24649 AND AN 504	SHEET 2 OF

DD FORM 1 SEP 57 672-1 (Coordinated)

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